

Prajwal Sathyanarayana

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SKILLS

Languages: Python, SQL, C, Java, C++, JavaScript

AI & LLM Engineering: Large Language Models (LLMs), RAG Pipelines, Agentic Workflows & Orchestration, Prompt Engineering, Tool Calling / Function Calling, Multimodal AI, NLP, Embeddings, Vector Databases (Pinecone, Weaviate), LangChain, LlamaIndex, Hugging Face, Model Evaluation & Eval Frameworks, LLMOps / MLOps

ML Libraries & Frameworks: PyTorch, TensorFlow, Scikit-Learn, LightGBM, XGBoost, Pandas, NumPy, FastAPI

Cloud: AWS (SageMaker, Lambda, RDS), Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, PostgreSQL, REST APIs

Observability & Evaluation: Model Monitoring, Drift Detection, Evaluation Harnesses, Automated Testing Pipelines, Production AI Reliability

EXPERIENCE

AI Engineer

Jan 2026 – May 2026

Axio Education

Scottsdale, AZ

- Built a production-grade document AI system using multimodal LLM (Gemini 1.5 Vision), implementing end-to-end LLM orchestration across 4 document modalities via an async FastAPI service, achieving a 78x speedup over the manual grading baseline at 9.2s median latency.
- Designed and implemented a Gemini-powered modality classification layer that auto-routes incoming PDF submissions across four processing pipelines (typed text, handwritten, multilingual, diagrams/tables), achieving greater than 85% routing accuracy and eliminating manual triage entirely.

Data Scientist

Aug 2025 – Dec 2025

Banner Health

Tucson, AZ

- Engineered scalable ETL workflows with PySpark, SQL, Databricks, AWS to centralize CMS Medicare and ACS SDOH data, creating a unified analytics-ready healthcare repository.
- Transformed and validated 1M+ records, improving data quality and reducing manual processing by 40%, while delivering automated county-level population health metrics and operational PowerBI dashboards.

Graduate Student Researcher

Jan 2025 – Jul 2025

University of Arizona - ACT Lab

Tucson, AZ

- Developed and optimized a RL workflow for robotic simulation, generating training datasets, troubleshooting complex Python/Linux environments, and analyzing 100K+ simulation steps to evaluate agent learning and performance metrics.
- Built LLM-adjacent NLP and data analysis pipelines in Python to debug, compare variants, and analyze real interaction logs from high-frequency simulations, improving experiment turnaround time by 30%.

Data Scientist

Jul 2022 – Jul 2024

Cognizant

Bangalore, India

- Architected scalable SQL and Python ETL pipelines across multi-source financial datasets for 3+ client portfolios, reducing cloud compute and data preparation costs by 25% while improving data quality and pipeline reliability.
- Designed and deployed supervised ML models (Random Forest, Gradient Boosting, XGBoost) for financial risk prediction, improving false-positive detection accuracy by 18% and reducing financial risk exposure through rigorous feature engineering.
- Automated production ML inference pipelines via AWS SageMaker with MLOps best practices, cutting model inference latency by 30%, lowering cloud infrastructure spend by 20%, and maintaining 99%+ uptime SLAs across production workloads.
- Designed agentic analytical workflows with tool-calling logic for multi-unit revenue forecasting, reducing operational forecasting costs by 40%; built automated Python dashboards improving operational efficiency by 35%.

PROJECTS

Fare Estimation Pipeline – Manhattan NYC TLC Data | *PostgreSQL, Python, FastAPI*

Aug 2025 – Dec 2025

- Designed PostgreSQL schemas and Python ETL pipelines to extract, transform, and load 3 years of NYC TLC trip data, automating ingestion and deployment via a cloud-native FastAPI service with data quality checks at each stage.

Predictive Modeling for Financial Markets | *Python, Regression*

Feb 2025 – Jul 2025

- Developed regression-based forecasting models on historical stock market and commodity datasets, building automated Python pipelines for data ingestion, preprocessing, validation, and feature engineering across multiple market segments.

EDUCATION

Master of Science in Data Science

University of Arizona

Bachelor of Engineering in Computer Science

Dayananda Sagar University